Gianluca Covini

PhD Student in Statistics and Computer Science, Bocconi University

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gianlucacovini

My current research interests lie in ***optimal transport****, with a focus on* entropic OT, Sinkhorn-type geometries and gradient-flow perspectives, *motivated by applications in statistics, optimization, and machine learning.*

Education

2025 – 2030

PhD in Statistics and Computer Science, *Bocconi University*, Milan, Italy

- **Advisor:** Prof. Hugo Lavenant.
- **Topic:** Variational problems in Sinkhorn geometry.
- **MPhil coursework** (*GPA: A*): Measure Theory; Probability Theory I–II; Statistical Theory I–II (extra); Bayesian Statistics I; Statistical Mechanics; Discrete Stochastic Processes (extra); Stochastic Differential Equations; Optimal Transport; Optimization; Graph Theory; Theoretical Computer Science; Programming in Julia; Deep Learning.

2022 – 2025

Master's Degree in Mathematics, *University of Pavia*, Pavia, Italy, *110/110* cum laude*; GPA: 29.5/30*

- **Relevant coursework:** Functional Analysis; Advanced Probability; Stochastic Processes; Foundations of Geometry; Mathematical Finance; Econometrics; Machine Learning; Finite Element Method; Dynamical Systems; Biomathematics.
- **Additional coursework attended beyond the curriculum:** Advanced Functional Analysis; Evolution Equations; Calculus of Variations.
- **Master's thesis:** *Dynamic Parameter Policies for LeadingOnes on Enhanced State Spaces*, supervised by Prof. Carola Doerr (Sorbonne Université/CNRS) and Prof. Stefano Gualandi (University of Pavia).

2022 – 2024

Alumnus of Merit-Based Program, *Almo Collegio Borromeo*, Pavia, Italy

- **Courses attended:** Optimal Transport for Optimization; Fuzzy Logic; Introduction to Stochastic Differential Equations; Quantitative Methods for Art.
- **Co-organized courses:** Machine Learning for Healthcare Management; Seminar Series in Mathematics.
- **Reading groups:** Hausdorff Measure and Fractals; Laplacian Eigenvalues, supervised by Prof. Dario Mazzoleni.

2019 – 2022

Bachelor's Degree in Mathematics, *University of Pavia*, Pavia, Italy, *110/110* cum laude*; GPA: 29.8/30*

- **Relevant coursework:** Mathematical Analysis; Linear Algebra; Algebra; Geometry; Probability; Mathematical Statistics; Numerical Analysis; Optimization; Foundations of Data Analysis; Programming.
- **Bachelor's thesis:** *Ewens Sampling Formula and its Applications to the Study of Population Biodiversity*, supervised by Prof. Emanuele Dolera.

Current Research Projects

- *Sinkhorn Propagation on Graphs*, joint with Hugo Lavenant (Bocconi University). Variational formulation of propagation problems on graphs of probability measures using Sinkhorn geometry and gradient flow perspective. Ongoing project.
- *Bayes Rule via Entropic Approximation of the Knothe–Rosenblatt Rearrangement*, joint with Carlo Ciliberto (UCL) and Giulia Luise (UCL/Microsoft Research). Study of entropic transport approximations of Knothe–Rosenblatt transport for Bayes update in filtering. Ongoing project.

Publications and Preprints

[1] Gianluca Covini, Denis Antipov, Carola Doerr. *Enhancing Parameter Control Policies with State Information*. Proceedings of the Foundations of Genetic Algorithms (FOGA 2025), Leiden, Netherlands, 2025. arXiv:2507.08368.

Undergraduate Research Experience

10/2024 – 12/2024 **Survey-Based Study in Optimization Under Uncertainty**, *Remote collaboration*

- Informal collaboration on *submodular functions for distributionally robust optimization*.
- **Supervisors:** Prof. Angelos Georghiou (University of Cyprus) and Prof. Rosario Paradiso (VU Amsterdam).
- **Outcome:** survey report.

07/2024 – 08/2024 **Statistics Summer Research Intern**, *Statistical Laboratory, University of Cambridge*, Cambridge, UK

- Worked on a simulation study of the statistical properties of stochastic optimization problems.
- **Supervisors:** Prof. Qingyuan Zhao and Tobias Freidling.
- **Outcome:** research report.

03/2024 – 09/2024 **Research Intern in Operations Research**, *LIP6 – Sorbonne Université/CNRS*, Paris, France

- Conducted research on dynamic algorithm configuration, developing optimal parameter policies for genetic algorithms in Python.
- **Supervisor:** Prof. Carola Doerr.
- **Outcomes:** Master's thesis; conference paper.

08/2023 – 03/2024 **Machine Learning Researcher**, *Sphaera*, Hybrid

- Developed a tracking algorithm for 5-a-side soccer using Python, NumPy, PyTorch, OpenCV, and Matplotlib.
- **Supervisors:** Dr. Mirko Messori and Dr. Giuseppe Roberto Marseglia.
- **Outcome:** research report; the algorithm developed remained confidential.

Talks and Presentations

Feb 2025 **Conference Presentation**, *ROADEF 2025 – Conference of the French Operations Research Society*, Champs-sur-Marne, France

Presented master's thesis research on dynamic parameter policies for randomized local search on the LeadingOnes problem. Abstract published in the ROADEF 2025 Book of Abstracts.

Jan 2025 **Seminar Talk**, *WIAS Berlin, Seminar Series "Modern Methods in Applied Stochastics and Nonparametric Statistics"*, Online

Presented the paper by Lambert et al., *Variational Inference via Wasserstein Gradient Flows*, in a public seminar of the WIAS research group Stochastic Algorithms and Nonparametric Statistics as part of a PhD interview process.

Teaching Experience

10/2021 – 09/2025 **Academic Tutor**, *University of Pavia*, Pavia, Italy

Teaching assistant for Calculus in the Engineering Bachelor's program and Probability in the Artificial Intelligence and Mathematics Bachelor's programs. Supervisors: Prof. Elisabetta Rocca, Prof. Abramo Agosti, Prof. Emanuele Dolera, and Prof. Carlo Orrieri.

10/2023 – 12/2023 **Lecturer – Advanced Computational Statistics**, *Almo Collegio Borromeo*, Pavia, Italy

Designed and taught the course *Advanced Computational Statistics*, covering theory and applications to real-world data in R. Supervisor: Prof. Emanuele Dolera.

Summer Schools and Short Courses

May 2026 **Mathematical Foundations of Machine Learning**, *CIRM*, Vason (TN), Italy

Dynamics, geometry and theoretical guarantees for neural networks, under an analysis point of view.

Jul 2025 **Summer School in Analysis and Machine Learning**, *Festum Pi*, Chania, Greece

Kinetic theory, stability of optimal transport, and diffusion models.

Jun 2025 **Theoretical Foundations of Machine Learning**, *MaLGa Center, University of Genoa*, Genoa, Italy

Statistical learning theory, kernel methods, neural networks, optimization, RKHS, convex analysis, and generalization.

Apr 2025 **Bocconi–StatML Spring School**, *University of Oxford*, Windsor, UK

Computational optimal transport and statistical learning with missing values; lecturers included Julie Josse and Gabriel Peyré.

Dec 2024 **PhD Winter School on Advanced Stochastic Optimization**, *NTNU*, Trondheim, Norway

Mixed-integer stochastic programming, decision making under uncertainty, distributionally robust optimization, and stochastic dual dynamic programming.

Jun 2024 **2nd Copenhagen PhD School of Stochastic Programming**, *University of Copenhagen*, Copenhagen, Denmark

Stochastic programming, multi-stage models, bounding techniques, scenario generation, and decision-dependent uncertainty.

Sep 2020 **Intensive School in Data Science**, *ISAGS, University of Pavia*, Pavia, Italy

Machine learning and applications in chemistry, genomics, and neuroscience.

Scholarships, Honors and Awards

2025 **Full PhD Scholarship**, Bocconi University.

2024–2025 **Selected funded participation** in advanced schools and workshops, including Mathematical Foundations of Machine Learning by CIRM, StatML – Bocconi Spring School, Festum Pi, Applied Mathematics School at KAUST, PhD School in Stochastic Programming at the University of Copenhagen, and School of Brain Cells & Circuits "Camillo Golgi" in Erice.

2024 **Merit-Based Scholarship**, Associazione Alunni Almo Collegio Borromeo.

2024 **Fully funded research stay**, Corpus Christi College, University of Cambridge.

2024 **Erasmus+ Traineeship Grant** for research internship at LIP6, Sorbonne Université/CNRS, Paris.

2024 **Research residence**, Maison de l'Italie, Cité Internationale Universitaire de Paris.

Certifications

Oct 2024 **TOEFL iBT 106/120 (C1)**, ETS.

Oct 2024 **GRE General Test**, ETS: Quantitative 168, Verbal 162.

Additional certifications: Getting Started with Deep Learning (NVIDIA, 2023); IBM Data Science (Coursera, 2021); DELF B1 (Ministère de l'Éducation Nationale, 2016).

Languages

Italian	Native	
English	C1	*TOEFL iBT 106/120.*
French	B1	*DELF B1.*

Computer Skills

Programming	Python, Julia, R, MATLAB	Scientific computing	NumPy, Pandas, PyTorch, OpenCV, Matplotlib
Optimization	Pyomo, Gurobi, CPLEX	Tools	Git, LaTeX, Linux, basic HPC usage
Other	Multiprocessing, Excel, WordPress	Cloud	Basic AWS usage

Extracurricular Activities

2022–2023 **Auditor & Data Analyst**, *JE Italy*
Monitored and analyzed confederation data; provided audit consulting services for junior enterprises.

2021–2023 **IT Consultant**, *JECO Pavia*, Pavia, Italy
Data management and digitalization projects for SMEs. As *Head of IT* (Sep 2021 – Oct 2022), managed area members and annual strategy planning using OKRs.

2022–2022 **Volunteer**, *Sant'Egidio – Youth for Peace*, Pavia, Italy
Organized recreational activities for children in difficult situations and activities aimed at countering school dropout.

2020
2021

Activity Manager, *Entrepreneurship Club Pavia*, Pavia, Italy
Organized public events with entrepreneurs and start-up founders; managed internal activities.

2018
2021

Co-Founder and Board Member, *The Most Maiorum*, Pavia, Italy
Organized cultural and charity events with hundreds of participants.

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